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Financial Dependence and Innovation: The Case of Public versus Private Firms

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1 Big picture

- **Question.** Does public listing help or hurt innovation? The paper’s answer is *it depends on whether the firm needs external capital*. The central comparison is between firms in industries that are externally finance dependent and firms in industries that are internally finance dependent.
- **Core tension.** Theory is ambiguous. Public markets can relax financing constraints, lower the cost of capital, and make risky innovative projects feasible. But public markets can also generate short-term pressure through myopic investors, analyst earnings targets, and stock-price-based managerial incentives.
- **Setting.** The authors study U.S. private and public nonfinancial firms from 1994 to 2004, using Standard & Poor’s Capital IQ matched to the NBER patent citation database.
- **Main empirical object.** The paper compares innovation input (R&D) and innovation output (patents, citations, originality, generality) across private and public firms, while allowing the public-listing effect to vary with industry external finance dependence.
- **Main finding.** Public firms in EFD industries spend more on R&D and produce larger, higher-quality, and more novel patent portfolios than comparable private firms. Public firms in IFD industries do *not* have a better innovation profile than private firms.
- **Identification logic.** The paper first uses industry-size matching and treatment-effect models to address self-selection into public status. It then uses a fuzzy regression discontinuity design around Nasdaq delisting thresholds to obtain quasi-experimental evidence on the effect of losing listing status.
- **Mechanism.** The financing benefits of public markets matter most when firms need outside capital. By contrast, short-termism from competition, analyst pressure, and real earnings management appears more relevant for firms that are not strongly dependent on external finance.
- **Why this paper matters.** The contribution is not “public is good” or “public is bad.” The contribution is the heterogeneity result: the innovation effect of stock-market access depends on financial dependence. That is the paper’s key idea and its most useful takeaway.

2 Data and measurement (key objects)

2.1 Sample construction

- **Innovation data.** Patent counts and citations come from the NBER Patent Citation Database, covering all patents granted by the USPTO from 1976 to 2006.
- **Financial data.** Firm-level accounting data come from S&P Capital IQ for 1994–2004.
- **Why the sample ends in 2004.** Patent applications take about two to three years to be granted, so stopping in 2004 allows the authors to observe subsequent patent outcomes with the 2006 patent database.
- **Listing-status correction.** Capital IQ codes firms by most recent listing status, so the authors manually reclassify firms using IPO dates and delisting information from Compustat, Thomson One, Jay Ritter’s IPO database, and CRSP first trading dates.

- **Exclusions.** Financials, utilities, firms without SIC codes, firms with missing total assets, firms with negative revenue, and firm-years with assets below \$5 million are excluded.
- **Coverage.** The full sample contains 2,392 private firms and 8,863 public firms.
- **Patent matching.** Financial data are merged to the patent database using GVKEY when possible and by company names otherwise, with manual checks to verify ambiguous matches.
- **Selection precaution.** Firm-years with no patent or citation information are assigned zeros so the innovation analysis is not restricted only to patenting firms.

2.2 Matched sample

- **Why matching is needed.** Capital IQ’s private-firm coverage is not a random sample of all private firms; it overrepresents larger private firms. The authors therefore build a matched sample to reduce cross-sectional differences.
- **Matching rule.** Each private firm is matched to a public firm in the same three-digit SIC industry that is closest in total assets in the first sample year.
- **Resulting sample.** This procedure produces 2,214 matched pairs of private and public firms.
- **Visual check.** Figure 1 shows that the size distributions of matched public and private firms overlap almost perfectly by construction.

2.3 Innovation measures

- **Input measure.** Innovation input is measured by R&D spending.
- **Output quantity.** Patent applications filed in year t are used rather than grants, because application year is closer to the timing of invention.
- **Output quality.** Patent citations are used to proxy for technological and economic importance.
- **Citation adjustments.** Citation counts are corrected for truncation bias using the Hall-Jaffe-Trajtenberg diffusion approach, and the paper also scales citations by year and technology-class averages so that citation inflation over time does not mechanically drive results.
- **Novelty.** The paper uses originality and generality to capture how broad the technological sources and uses of a patent are.

2.4 External finance dependence and innovation intensity

- **Firm-level need for external finance.** Following Rajan and Zingales (1998), a firm’s need for external finance in a year is measured as the fraction of capital expenditures not financed by internal cash flow.
- **Industry-level EFD index.** For each two-digit SIC industry and year, the authors compute the median firm-level external-finance need. An industry’s EFD index is the percentile rank of the time-series median over 1994–2004.
- **Classification.** Industries with positive external-finance dependence are classified as EFD; those with negative dependence are classified as IFD.

- **Innovation-intensity index.** The paper constructs a patent-based industry innovation-intensity index as the percentile rank of the industry’s time-series median patenting among patent-producing firms. It also builds an alternative index using median R&D for firms with positive R&D.
- **Why this matters.** The authors want to separate two ideas that are easy to confound: industries may differ in *how innovative* they are and in *how dependent on outside finance* they are. Figure 2 shows these are not the same thing.

3 Models and empirical specifications (all equations + interpretation)

3.1 Patent novelty measures

The paper’s first key formulas define patent originality and generality.

$$\text{Originality}_i = 1 - \sum_j F_{ij}^2, \quad (1)$$

where F_{ij} is the share of patents cited by patent i that belong to class j .

- **Interpretation.** A patent is more original if it draws on a broader set of prior technologies rather than a narrow single class.
- **Economic meaning.** This is a measure of novelty on the *input* side of invention.

$$\text{Generality}_i = 1 - \sum_j G_{ij}^2, \quad (2)$$

where G_{ij} is the share of future citing patents that belong to class j .

- **Interpretation.** A patent is more general if future innovators from many different technological classes cite it.
- **Economic meaning.** This is a measure of novelty and breadth on the *output* side of invention.

3.2 Baseline panel specification

The main comparison of innovation outcomes across private and public firms is:

$$Y_{i,k,t} = \alpha + \beta \text{Public}_{i,k,t} + \gamma X_{i,k,t-1} + \eta_k + \zeta_t + \varepsilon_{i,k,t}, \quad (3)$$

where $Y_{i,k,t}$ is alternately $\ln(1 + \text{R\&D})$, $\ln(1 + \text{Patent})$, $\ln(1 + \text{Citations})$, originality, or generality.

- **Regressor of interest.** $\text{Public}_{i,k,t} = 1$ for public firms and 0 for private firms.
- **Controls.** $X_{i,k,t-1}$ includes log sales, tangibility, cash, age, capital expenditure, sales growth, and ROA.
- **Fixed effects.** η_k are industry fixed effects and ζ_t are year fixed effects.

- **Key coefficient.** β is the difference in innovation between public and private firms within the same industry-year after conditioning on observables.
- **Prediction.** The sign of β is ambiguous ex ante because financing benefits and market short-termism work in opposite directions.

3.3 Treatment-effect / selection model

Because firms do not randomly choose public status, the paper uses a treatment-effect model. The latent-selection equation is

$$\text{Public}_i = \begin{cases} 1 & \text{if } \text{Public}_i^* > 0, \\ 0 & \text{if } \text{Public}_i^* \leq 0, \end{cases} \quad \text{Public}_i^* = \pi + \delta Z_i + v_i. \quad (4)$$

- **Interpretation.** Z_i contains variables that affect the firm's decision to go public, and the first-stage probit delivers an inverse Mills ratio that is added to the outcome equation.
- **What this buys the authors.** This is a Heckman-style correction for endogenous selection into public status.
- **Important nuance.** The baseline treatment-effect model does not use a classical external instrument for public status; its identifying variation comes from functional-form assumptions and the nonlinearity of the inverse Mills ratio.

3.4 Heterogeneity by external finance dependence

The paper's central heterogeneity regression is

$$Y_{i,k,t} = \alpha + \beta \text{Public}_i + \delta \text{EFD}_{i,k} + \theta (\text{Public}_i \times \text{EFD}_{i,k}) + \gamma X_{i,k,t-1} + \lambda (X_{i,k,t-1} \times \text{EFD}_{i,k}) + \phi_i + \varepsilon_{i,k,t}. \quad (5)$$

- **Key coefficient.** θ is the main coefficient in the entire paper.
- **Interpretation.** If $\theta > 0$, the innovation effect of being public is more positive in industries that depend more on external finance.
- **Plain-English reading.** This is the direct empirical implementation of the paper's main claim that stock-market access matters most when innovative firms need outside capital.

3.5 Fuzzy regression discontinuity around Nasdaq delisting

To obtain cleaner causal evidence, the paper studies firms near Nasdaq's continued-listing thresholds. Let c_i denote the forcing variable and c_0 the threshold.

$$P(\text{Delisting}_i = 1 \mid c_i) = \begin{cases} f_1(c_i) & \text{if } c_i < c_0, \\ f_0(c_i) & \text{if } c_i \geq c_0, \end{cases} \quad (6)$$

with $f_1(c_0) \neq f_0(c_0)$.

- **Forcing variable construction.** The authors normalize the bid-price requirement and three “core” listing criteria (net tangible assets / shareholder equity, market capitalization, net income). They then take the minimum of the normalized bid-price variable and the maximum of the three normalized core variables.
- **Threshold.** The normalized threshold is zero; falling below zero means the firm violates Nasdaq’s continued-listing requirement.
- **Why fuzzy rather than sharp RD.** Crossing the threshold does not mechanically delist every firm, so treatment probability jumps but does not go from 0 to 1.

The treatment probability is written as

$$\mathbb{E}[\text{Delisting}_i \mid c_i] = P(\text{Delisting}_i = 1 \mid c_i) = f_0(c_i) + [f_1(c_i) - f_0(c_i)]z_i, \quad (7)$$

where $z_i = 1(c_i \leq c_0)$ indicates the side of the threshold.

Assuming a polynomial in the forcing variable, the first-stage can be written as

$$\mathbb{E}[\text{Delisting}_i \mid c_i] = \gamma_0 + \gamma_1 c_i + \gamma_2 c_i^2 + \cdots + \gamma_p c_i^p + \lambda z_i + \delta_1 c_i z_i + \delta_2 c_i^2 z_i + \cdots + \delta_p c_i^p z_i. \quad (8)$$

The reduced-form outcome equation under the simplest linear specification is

$$Y_{i,t+N} = \alpha + \beta_1 z_{i,t} + \beta_2 c_{i,t} + \beta_3 X_{i,t} + \varepsilon_{i,t}, \quad (9)$$

where $Y_{i,t+N}$ is future innovation in years $t + 1$, $t + 2$, or $t + 3$.

- **Interpretation.** $z_{i,t}$ is the quasi-experimental shock. Firms just below and just above the threshold are plausibly similar except for a discrete change in delisting probability.
- **Controls.** $X_{i,t}$ includes log assets, tangibility, cash, age, capex, and ROA.
- **Design scope.** The RD identifies the effect of *losing public status* for firms near the delisting threshold, not the effect of public listing for every firm in the economy.

The fuzzy-RD estimand is

$$\beta = \frac{\lim_{c \rightarrow c_0^+} \mathbb{E}[Y_i \mid c_i] - \lim_{c \rightarrow c_0^-} \mathbb{E}[Y_i \mid c_i]}{\lim_{c \rightarrow c_0^+} \mathbb{E}[\text{Delisting}_i \mid c_i] - \lim_{c \rightarrow c_0^-} \mathbb{E}[\text{Delisting}_i \mid c_i]}. \quad (10)$$

- **Interpretation.** This is the local average treatment effect of delisting for compliers near the threshold.

3.6 Real earnings management construction

To study short-termism, the paper measures abnormal discretionary spending using the Roychowdhury-style cross-sectional regression

$$\frac{\text{DISX}_{i,t}}{TA_{i,t-1}} = \alpha + \beta_1 \left(\frac{1}{TA_{i,t-1}} \right) + \beta_2 \left(\frac{\text{Sales}_{i,t-1}}{TA_{i,t-1}} \right) + \varepsilon_{i,t}. \quad (11)$$

- **Construction.** DISX is discretionary expenditure, including advertising and selling, general, and administrative expenses.
- **Interpretation.** The fitted value is the “normal” level of discretionary spending; abnormal discretionary spending is the difference between normal and actual. A higher REM measure means the firm is cutting discretionary outlays more aggressively to hit short-term earnings targets.

The paper then estimates

$$\begin{aligned}
\text{REM}_{i,k,t} = & \alpha + \beta_1 \text{Intensity}_{i,k} + \beta_2 \text{EFD}_{i,k} + \beta_3 (\text{Intensity}_{i,k} \times \text{EFD}_{i,k}) \\
& + \beta_4 \text{MVE}_{i,k,t-1} + \beta_5 \text{MB}_{i,k,t-1} + \beta_6 \text{ROA}_{i,k,t} \\
& + \beta_7 (\text{MVE}_{i,k,t-1} \times \text{EFD}_{i,k}) + \beta_8 (\text{MB}_{i,k,t-1} \times \text{EFD}_{i,k}) \\
& + \beta_9 (\text{ROA}_{i,k,t} \times \text{EFD}_{i,k}) + \zeta_t + \zeta_t \times \text{EFD}_{i,k} + \varepsilon_{i,k,t}.
\end{aligned} \tag{12}$$

- **Purpose.** This mechanism regression asks whether innovative public firms in EFD industries are more or less likely to engage in real earnings management than innovative public firms in IFD industries.

3.7 Analyst coverage and future innovation

The analyst-pressure specification is

$$Y_{i,k,t+3} = \alpha + \beta \ln(\text{Coverage})_{i,k,t} + \theta \text{EFD}_{i,k} \times \ln(\text{Coverage})_{i,k,t} + \gamma X_{i,k,t-1} + \lambda X_{i,k,t-1} \times \text{EFD}_{i,k} + \xi_i + \zeta_t + \varepsilon_{i,k,t}. \tag{13}$$

- **Dependent variables.** Three-year-ahead patents, citations, originality, and generality.
- **Regressor of interest.** Analyst coverage is the log of one plus the average monthly number of earnings-forecast analysts from I/B/E/S.
- **Interpretation.** A negative β means analyst pressure reduces future innovation. A positive θ means this negative effect is weaker for firms in EFD industries.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- **Core comparison.** Compare public and private firms with similar industry and size, then control flexibly for firm characteristics.
- **Selection correction.** Use a treatment-effect model so the public/private comparison is not driven only by observable differences and simple OLS selection bias.
- **Strength.** The approach uses a very large private-firm sample, which is rare in innovation work.
- **Weakness.** The treatment-effect model still relies on selection-on-observables plus functional-form assumptions. It is helpful, but it is not the strongest causal design in the paper.

4.2 Fuzzy RD design around Nasdaq delisting

- **Treatment.** Losing public listing status through Nasdaq delisting.
- **Instrument-like source of variation.** The threshold indicator $z_i = 1(c_i \leq 0)$ creates a discontinuous change in the probability of delisting and functions as the instrument in the fuzzy-RD 2SLS setup.
- **Why relevance holds.** Figure 6 shows a sharp jump in delisting probability at the threshold.
- **Why exclusion is plausible.** Firms just above and below the threshold should be similar except for their increased chance of delisting, provided they cannot precisely manipulate the forcing variable.

4.3 Validity checks

- **Density test.** The McCrary test shows no strong evidence of manipulation of the forcing variable at the threshold; the estimated discontinuity in density is small (0.11 with standard error 0.11).
- **Balance test.** Near the cutoff, most firm characteristics are balanced, though tangibility, ROA, and age differ somewhat; the authors therefore include covariates in the RD estimation.
- **Placebos.** Using an artificial delisting threshold or artificial delisting year produces no discontinuity in innovation.
- **Interpretation caution.** The fuzzy-RD evidence is especially persuasive because it lines up with the broader matched-sample results: delisting hurts innovation in EFD industries but not in IFD industries.

5 Tables and figures

5.1 Figure 1: Size and age distribution of public and private firms

Read:

- The matched sample succeeds on size: public and private firms have almost identical size distributions.
- Age still differs somewhat, especially in IFD industries, with public firms tending to be older.
- This figure motivates why the paper keeps controlling for age even after size-industry matching.

5.2 Table 1: Characteristics and innovation activities of private and public firms

Read:

- Even in the matched sample, public firms do more R&D, produce more patents, receive more citations, and have higher originality and generality than private firms on average.
- The raw differences are large: in the matched sample, public firms have 3.70 patents versus 0.59 for private firms and 3.40 citations versus 1.59.

- This table is descriptive only. It motivates the question but cannot separately identify financing benefits from selection.

Table 1

Characteristics and innovation activities of private and public firms.

In this table, we report the means of characteristic variables for the full sample of private and public firms and for an industry-and-size matched sample. The full sample (Panel A) consists of 11,255 US firms from Standard and Poor's Capital IQ from 1984 to 2004. The matched sample (Panel B) has 2,214 matched pairs of private and public firms. $\ln(\text{Sales})$ is the log of total revenue. ΔGrowth is the first difference of natural logarithm of total revenue. Tangible is tangible (fixed) assets scaled by total assets. Cash is total cash scaled by total assets. ROA is EBITDA (earnings before interest, taxes, depreciation, and amortization) divided by total assets. Age is the difference between current year and founding year. Capex is capital expenditures scaled by total assets. $\ln(\text{R\&D})$ is natural logarithm of one plus research and development expenditures. Patent is the number of patent applications submitted by a firm in a given year. Citations is citations per patent adjusted for truncation bias by weighting the number of citations with an estimated distribution of citation-lag. Originality of Patent is Herfindahl index of cited patents, and Generality is Herfindahl index of citing patent. Tangible , Cash , ROA , and Capex are reported as percentages. Difference is the difference in means of private and public firms from the t-test.

Panel A: Full sample											
Listing status	$\ln(\text{Sales})$	S. Growth	Tangible	Cash	ROA	Age	Capex	$\ln(\text{R\&D})$	Patent	Citations	Originality
Private	4.55	0.27	29.74	14.66	2.67	26.21	7.20	0.50	1.02	2.84	0.38
Public	4.78	0.15	26.20	18.89	3.79	33.50	6.31	0.86	7.48	4.09	0.40
Difference	0.23	-0.12	-3.54	4.23	1.11	7.30	-0.89	0.36	6.46	1.25	0.02
t-statistic	9.86	-16.38	-15.27	18.18	4.41	20.57	-12.21	26.24	10.18	8.98	1.80
Panel B: Matched sample											
Listing status	$\ln(\text{Sales})$	S. Growth	Tangible	Cash	ROA	Age	Capex	$\ln(\text{R\&D})$	Patent	Citations	Originality
Private	4.81	0.21	31.28	11.64	5.78	28.96	6.76	0.37	0.59	1.59	0.55
Public	4.82	0.15	28.10	16.59	4.95	36.30	6.42	0.64	3.70	3.40	0.57
Difference	0.02	0.06	3.19	-4.95	0.84	-7.35	-0.34	-0.26	-3.11	-1.82	-0.03
t-stat	3.80	-6.13	-9.53	16.81	-2.59	14.29	-3.30	17.24	7.83	11.07	2.82

Table 2

External finance dependence and innovation.

This table reports the first-step estimation results of the treatment effect model for industry and size matched firms in all industries, as well as firms in external finance dependent (EFD) and internal finance dependent (IFD) industries (Panel A), the second-step estimation results for private and public firms in EFD (Panel B) and IFD (Panel C) industries, and the differential effects of public listing on innovation between EFD and IFD industries (Panel D-Panel G). The treatment effect model is estimated with a two-step approach. In the first step, we estimate the probability of being public based on a firm's logarithm of sales, capital expenditure, growth in sales, return on assets (ROA), leverage, and innovation intensity index from a probit model. The inverse Mills ratio (Mills) is included in the second step to adjust for selection bias. The dependent variable is the measures of innovation activities: the natural logarithm of one plus R&D, the natural logarithm of one plus the number of patents, the natural logarithm of one plus truncation bias-adjusted citations, originality, and generality. Public is a dummy variable equal to one for public firms and zero for private firms. The control variables include $\ln(\text{Assets})$, Tangible , Cash , Age , Capex , growth in sales, and ROA. Year and industry fixed effects are controlled. In Panel D-Panel G, a treatment effect model with the second-step estimation as following is estimated: $Y_{i,t} = \alpha + \beta \text{Public}_i + \delta \text{EFD}_{i,t} + \theta \text{Public}_i \times \text{EFD}_{i,t} + \gamma X_{i,t-1} + \lambda X_{i,t-1} \times \text{EFD}_{i,t} + \phi \text{Mills}_i + \epsilon_{i,t}$, where $\text{EFD}_{i,t}$ is an industry external finance index and $X_{i,t-1}$ are the control variables. The model is estimated for four samples: industry-and-size matched private and public firms (Panel D), age, year, and size matched pairs of private and public firms in EFD and IFD industries (Panel E), age-year-and-R&D matched pairs of private and public firms in EFD and IFD industries (Panel F), and the sample excluding acquirers in the merger and acquisition (M&A) transactions (Panel G). The coefficients on the control variables are not reported. N is the number of observations. Two-step consistent standard errors are reported in the brackets. ***, **, and * indicate the 1%, 5%, and 10% significance level, respectively.

Panel A: Treatment effect model first step					
Variable	All	EFD industries	IFD industries		
Capex	0.7917*** [0.1971]	0.9105*** [0.2083]	0.9592 [0.6897]		
S.Growth	-0.0646** [0.0266]	-0.0882*** [0.0286]	0.0904 [0.0807]		
ROA	-0.5451*** [0.0858]	-0.6769*** [0.0915]	0.1305 [0.2625]		
Sales	-0.0023 [0.0071]	0.0059 [0.0077]	-0.0681*** [0.0188]		
Leverage	-1.6542*** [0.0430]	-1.5700*** [0.0473]	-1.5700*** [0.1057]		
Intensity	0.1570*** [0.0510]	0.3182*** [0.0598]	0.0625 [0.1029]		
EFD	0.1679*** [0.0542]				
N	12,253	10,137	2,116		
Variable	$\ln(\text{R\&D})$	$\ln(\text{Patent})$	$\ln(\text{Citations})$	Originality	Generality
Panel B: External finance dependent industries					
Public	0.7535*** [0.0699]	0.5072*** [0.0511]	0.5109*** [0.0589]	0.1301*** [0.0371]	0.1709*** [0.0522]
Mills	-0.3759*** [0.0421]	-0.1745*** [0.0309]	-0.1812*** [0.0356]	-0.0315 [0.0207]	-0.0623** [0.0280]
N	10,137	10,137	10,137	2,480	2,480
Panel C: Internal finance dependent industries					
Public	0.0538 [0.0602]	-0.0844 [0.0644]	-0.0911 [0.0902]	-0.0495 [0.1212]	-0.2141 [0.1540]
Mills	-0.0180 [0.0370]	0.0767 [0.0395]	0.0832 [0.0554]	0.0487 [0.0734]	0.0891 [0.0930]
N	2,116	2,116	2,116	192	192
Panel D: Industry and size matched sample					
EFD × Public	0.4996*** [0.0845]	0.5538*** [0.0633]	0.5340*** [0.0746]	0.2422*** [0.0608]	0.1122 [0.0906]
N	12,253	12,253	12,253	2,672	2,672
Panel E: Age, year, and size matched EFD and IFD pairs					
EFD × Public	0.2663** [0.1437]	0.3929*** [0.0740]	0.4642*** [0.0976]	0.2485*** [0.1215]	0.1202 [0.1830]
N	3,531	3,531	3,531	525	525
Panel F: Age, year, and R&D matched EFD and IFD pairs					
EFD × Public	0.9247*** [0.1094]	0.4452*** [0.0873]	0.4207*** [0.0914]	0.2035 [0.2033]	0.3482* [0.2104]
N	2,533	2,533	2,533	356	356
Panel G: Exclude M&As					
EFD × Public	0.1741* [0.0892]	0.3429*** [0.0640]	0.4533*** [0.0846]	0.3315*** [0.0701]	0.2113** [0.1015]
N	8,703	8,703	8,703	1,715	1,715

5.3 Table 2: External finance dependence and innovation

Read:

- This is the core table. In the first-stage selection equation, firms in EFD industries and more innovation-intensive industries are more likely to go public.
- In EFD industries, the public-firm coefficient is positive and highly significant for R&D, patents, citations, originality, and generality. In IFD industries, the same coefficients are insignificant.
- The interaction $\text{EFD} \times \text{Public}$ is positive and significant for most innovation measures. That is the paper's main empirical result.
- The result survives more demanding pair-matching exercises and survives the exclusion of acquirers in M&A transactions.

5.4 Figure 2: Innovation intensity index and EFD index

Read:

- Industries that rely more on external finance are not systematically the industries in which innovation matters most.
- The paper reports a small and insignificant correlation between the innovation-intensity index and the EFD index (about 0.08).
- This figure is important because it blocks a simple alternative story that the EFD results are really just an “innovation-intensive industries” result.

5.5 Figure 3: Number of IPOs

Read:

- IPO activity is concentrated in EFD industries and in high-innovation-intensity industries.
- The concentration is especially strong in the late-1990s and 2000 wave.
- This figure visually supports the financing-benefits channel: firms that need outside capital and have important innovative opportunities are precisely the firms most likely to enter public equity markets.

5.6 Figure 4: Regression discontinuity — Nasdaq delisting and innovation

Read:

- For EFD firms, there is a clear downward jump in post-delisting R&D, patenting, and citations when the forcing variable falls below the Nasdaq threshold.
- For IFD firms, the same discontinuity is essentially absent.
- This figure is the visual causal backbone of the paper: losing listing status hurts innovation only where outside finance is important.

5.7 Figure 5: Placebo tests with artificial threshold and artificial delisting year

Read:

- When the authors impose fake thresholds or fake delisting dates, the discontinuities disappear.
- This is exactly what we want to see: the true break happens only at the economically meaningful Nasdaq threshold and actual delisting timing.

5.8 Figure 6 and Table 3: RD assumptions and local balance

Read:

- Figure 6 Panel A shows a jump in the probability of delisting at the threshold, establishing a strong first stage for fuzzy RD.

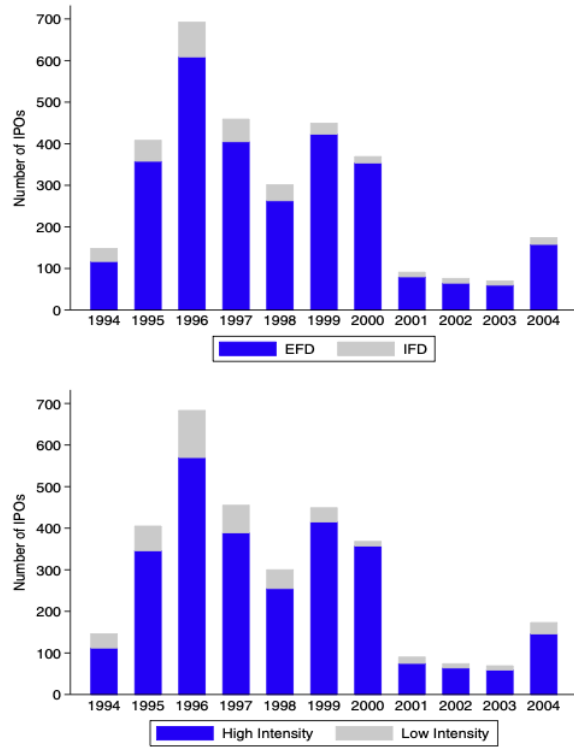


Fig. 3. Number of initial public offerings (IPOs). This figure presents the number of IPOs in external finance dependent (EFD) and internal finance dependent (IFD) industries, and in high and low innovation intensity industries over 1994–2004 for the sample firms. Industries with a positive (negative) value of EFD measure are regarded as external (internal) finance dependent. Industries with an innovation intensity index higher (lower) than the index median value are regarded as high (low) innovation intensity industries.

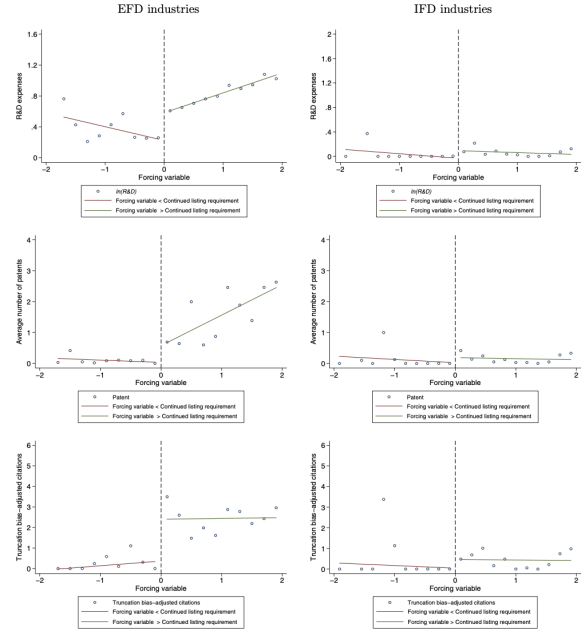


Fig. 4. Regression discontinuity: Nasdaq delisting and innovation. This figure shows the effect of Nasdaq delisting on innovation for firms in external finance dependent industries (EFD) and internal finance dependent (IFD) industries. We plot the average R&D expenditure, the average number of patents, and the average truncation bias-adjusted citation over the post-delisting period for Nasdaq delisted firms and over the pseudo post-delisted period for public firms on his width of 0.2. We construct the forcing variable by taking the minimum between the log normalized bid price and the maximum value of the three log normalized core Nasdaq continued listing requirements. Nasdaq continued listing requirement variables are normalized to have a value of zero at the threshold. Delisting occurs when the forcing variable falls below the threshold. The sample period is from 1997 to 2004.

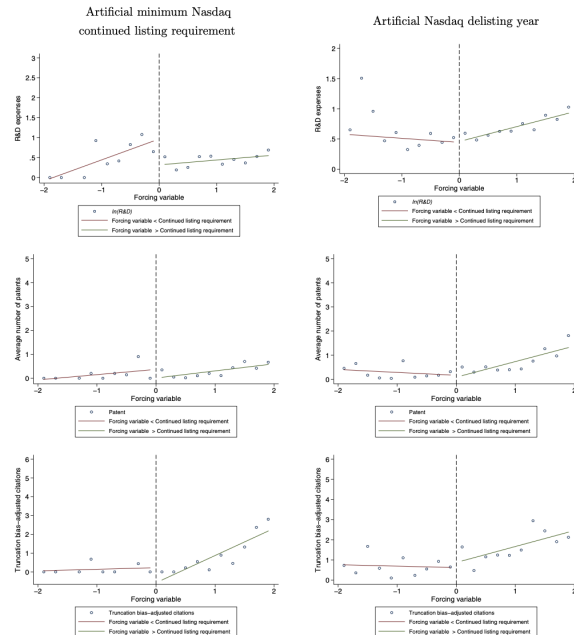
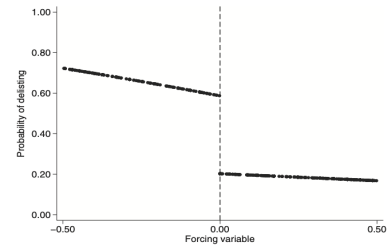


Fig. 5. Placebo test: artificial delisting threshold and artificial delisting year. This figure shows the placebo effect of a Nasdaq delisting on innovation using artificial minimum Nasdaq continued listing requirement and artificial Nasdaq delisting year. We plot the average R&D spending, the average number of patents, and the average truncation bias-adjusted citations for firms in external finance dependent industries. The sample period is from 1997 to 2004.

Panel A: Discontinuity of the probability of treatment (delisting) at the threshold



Panel B: Density of the forcing variable following McCrary (2008)

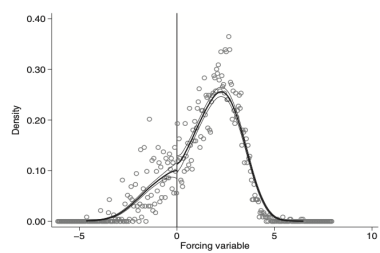


Fig. 6. Tests of regression discontinuity assumptions: delisting. This figure presents the discontinuity of the probability of treatment (delisting) at the threshold (Panel A) and the density of the forcing variable following McCrary (2008) (Panel B). In Panel A, we plot the probability of delisting as a function of the forcing variable. The forcing variable is measured as the minimum between the log normalized bid price and the maximum value of the three log normalized core Nasdaq continued listing requirements. The threshold is the minimum requirements for Nasdaq continued listing. Delisting occurs when the forcing variable falls below the threshold. In Panel B, the circles represent the density estimate and the solid line is the fitted function of the forcing variable with a 95% confidence interval.

Table 3

Covariates in the fuzzy regression discontinuity sample.

This table presents differences in characteristics of delisted and non-delisted firms near the Nasdaq continued listing requirements. The tests are conducted for the forcing variable within the interval of $[-0.1, +0.1]$. $\ln(\text{Assets})$ is the natural logarithm of total assets. $\ln(\text{Sales})$ is the natural logarithm of total revenue. $S.\text{Growth}$ is the first difference of the natural logarithm of total revenue. Tangible is tangible assets scaled by total assets. Cash is total cash scaled by total assets. ROA is EBITDA (earnings before interest, taxes, depreciation, and amortization) divided by total assets. Age is the difference between current year and founding year. Leverage is the ratio of total debt to total assets. Capex is capital expenditures scaled by total assets. Tangible , Cash , ROA , Leverage , and Capex are reported as percentages. Difference is the difference in medians of the two groups of firms. $p\text{-value}$ is the p -value of the Wilcoxon rank-sum test.

Delisting status	$\ln(\text{Assets})$	$\ln(\text{Sales})$	S. Growth	Tangible	Cash	ROA	Leverage	Age	Capex
Non-Delisted	3.12	3.34	0.02	15.64	5.09	-1.35	29.00	18.00	3.13
Delisted	3.02	3.30	0.05	10.25	7.87	-5.42	33.94	10.00	2.56
Difference	0.10	0.04	-0.03	5.39	-2.78	4.07	-4.94	8.00	0.57
p-value	0.54	0.90	0.78	0.02	0.12	0.01	0.57	0.01	0.19

Table 4

Fuzzy regression discontinuity estimation with covariates.

This table reports the results of the fuzzy RD estimation with covariates using the Nasdaq continued listing requirement for firms in external finance dependent (EFD) (Panel A) and internal finance dependent (IFD) industries (Panel B). The two-stage least squares (2SLS) approach is used to estimate the reduced form model $Y_{i,t} = \alpha + \beta_1 X_{i,t} + \beta_2 X_{i,t} + \beta_3 X_{i,t} + \epsilon_{i,t}$. The dependent variables are: the natural logarithm of one plus R&D, the natural logarithm of one plus the number of patents, the natural logarithm of one plus truncation bias-adjusted citations, originality, generality in the post-delisting years ($t+1, t+2, t+3$). The independent variable $X_{i,t}$ is an indicator variable that equals one if the forcing variable $z_{i,t}$ is less or equal to the threshold. We use the minimum between the log normalized bid price and the maximum value of the three log normalized core Nasdaq continued listing requirements as the forcing variable and the normalized minimum continued listing standard as the threshold for delisting from the Nasdaq. $X_{i,t}$ is a set of covariates that can influence innovation. The coefficient, β_1 , for treatment assignment is reported and 2SLS standard errors are reported in brackets. ***, **, and * indicate the 1%, 5%, and 10% significance level, respectively.

Year	$\ln(\text{R\&D})$	$\ln(\text{Patent})$	$\ln(\text{Citations})$	Originality	Generality	N
Panel A: EFD industries						
$t+1$	-0.3828*** [0.0854]	-0.3770*** [0.0646]	-0.7300*** [0.0989]	-0.1691* [0.0965]	-0.3998*** [0.1253]	3,247
$t+2$	-0.4666*** [0.1221]	-0.3722*** [0.0942]	-0.7521*** [0.1304]	-0.044 [0.1278]	-0.5027*** [0.1758]	2,795
$t+3$	-0.4512*** [0.1745]	-0.3863*** [0.1304]	-0.6488*** [0.1540]	-0.0307 [0.1197]	-0.2968* [0.1697]	2,464
Panel B: IFD industries						
$t+1$	-0.1384 [0.0978]	-0.0440 [0.0812]	-0.1340 [0.1435]	-0.1703 [0.5199]	0.2531 [0.7140]	440
$t+2$	-0.2135 [0.1398]	0.0044 [0.1410]	-0.0874 [0.2468]	0.8540 [0.7145]	-0.2882 [0.8207]	377
$t+3$	-0.6308*** [0.2328]	-0.1627 [0.2303]	-0.5316 [0.3497]	-0.2873 [0.3130]	-0.3300 [0.4322]	338

- Figure 6 Panel B shows no major bunching of the forcing variable at the threshold, supporting the no-precise-manipulation assumption.
- Table 3 shows local balance is broadly reasonable, although tangibility, ROA, and age differ somewhat; this is why covariates are added in the RD estimation.

5.9 Table 4: Fuzzy regression discontinuity estimation with covariates

Read:

- Delisting from Nasdaq significantly lowers future R&D, patents, and citations for firms in EFD industries in years $t+1$, $t+2$, and $t+3$.
- The paper highlights that by $t+3$, delisted EFD firms spend much less on R&D and generate about 47% fewer patents than comparable listed firms near the threshold.
- For IFD firms, the RD estimates are mostly insignificant. This sharp contrast is why the paper's heterogeneity result is so convincing.

5.10 Table 5: Real earnings management and innovation

Read:

- Public firms in IFD industries engage in more real earnings management than public firms in EFD industries.
- Within EFD industries, the most innovative firms do the *least* REM.
- The mechanism message is subtle: needing external capital does not mechanically make firms more myopic. If anything, innovative public EFD firms appear less willing to cut long-term spending to massage short-term earnings.

Table 5

Real earnings management and innovation.

This table reports the results for the relation between innovation and real earnings management (REM) for public firms. In Panel A, we compare the REM of firms in external finance dependent (EFD) and internal finance dependent (IFD) industries using both the matched and full samples. In Panel B, we classify public firms in EFD industries into three groups based on the innovation intensity index. REM is measured as the difference between the normal level of discretionary expenses and the actual discretionary expenses. We estimate the normal discretionary expenses by the fitted values from the following cross-sectional regression for every industry and year: $DISX_{it}/TA_{it-1} = \alpha + \beta_1(1/TA_{it-1}) + \beta_2(Sales_{it-1}/TA_{it-1}) + \epsilon_{it}$, where DISX is the discretionary expenditures, including advertising expenses and selling and general administrative expenses; TA is total assets; and Sales is total revenue. A higher value of REM indicates a higher degree of real earnings management. Difference is the difference in the average REM between public firms in EFD and IFD industries. Panel B reports the average REM of the three groups and the significance levels of difference from Group 1. In Panel C, Column 4, we report the estimation results of the regression model: $REM_{it} = \alpha + \beta_1 Intensity_{it} + \beta_2 EFD_{it} + \beta_3 Intensity_{it} \times EFD_{it} + \beta_4 MVE_{it-1} + \beta_5 MB_{it-1} + \beta_6 ROA_{it-1} + \beta_7 MVE_{it-1} \times EFD_{it} + \beta_8 MB_{it-1} \times EFD_{it} + \beta_9 ROA_{it-1} \times EFD_{it} + \epsilon_{it}$, where Intensity is the innovation intensity index; EFD is the external finance dependence dummy; MVE is the logarithm of market value of equity; MB is market-to-book ratio; and ROA is return on assets. Column 1 presents the estimation results of the model without Intensity and Intensity $\times$ EFD. Columns 2 and 3 present the estimation results for firms in EFD and IFD industries, respectively. The coefficients on the control variables are not reported. The robust standard errors are reported in brackets. ***, **, and * indicate the 1%, 5%, and 10% significance level, respectively.

Panel A: EFD versus IFD industries				
Industries	Matched sample	Full sample		
IFD industries	1.36	2.55		
EFD industries	-6.11	-1.45		
Difference	-7.47***	-4.01***		
Panel B: Innovative versus non-innovative firms in EFD industries				
Group	Matched sample	Full sample		
1 Least innovative	0.76	9.01		
2 Moderately innovative	-6.19***	-2.46***		
3 Most innovative	-15.94***	-12.25***		
Panel C: Real earnings management				
Variable	(1)	EFD only (2)	IFD only (3)	(4)
EFD	-19.5767*** [4.9566]			6.7438 [9.9592]
Innovation Intensity		-16.9262*** [1.2159]	2.7254 [7.3189]	2.7254 [7.2858]
Intensity $\times$ EFD				-19.6516*** [7.3867]
N	7,648	6,312	1,096	7,408

Table 6

Competition and innovation.

This table reports the estimation results for the differential effects of competition on innovation between competitive and noncompetitive industries. A treatment effect model with the second-step estimation as follows is estimated: $Y_{it} = \alpha + \beta Public_{it} - \gamma Competitive_{it} + \theta Public_{it} \times Competitive_{it} + \gamma X_{it-1} + \delta X_{it-1} \times Competitive_{it} + \delta MVE_{it-1} + \epsilon_{it}$, where Competitive_{it} is a dummy variable equal to one for competitive industries and zero for noncompetitive industries. We measure industry competitiveness using the Hoberg and Phillips Industry Concentration Measure (HPCM). An industry is classified as competitive if its HPCM falls into the bottom tercile of all industries and as non-competitive if its HPCM falls into the top tercile. X_{it-1} includes $\ln(Sales)$, Tangible, Cash, Age, Capex, growth in sales, and ROA. The model is estimated separately for firms in external finance dependent (EFD) and internal finance dependent (IFD) industries. The coefficients on the control variables are not reported. Two-step consistent standard errors are reported in brackets. ***, **, and * indicate the 1%, 5%, and 10% significance level, respectively.

Variable	ln(R&D)	ln(Patent)	ln(Citations)	Originality	Generality
Panel A: EFD industries					
Competitive $\times$ Public	0.5292*** [0.0823]	0.0053 [0.0533]	0.0518 [0.0526]	0.0234 [0.0446]	-0.0458 [0.0571]
N	26,501	26,501	26,501	8,678	8,678
Panel B: IFD industries					
Competitive $\times$ Public	-0.1837*** [0.0397]	-0.0863* [0.0489]	-0.1075** [0.0518]	-0.2776* [0.1658]	-0.2077 [0.2651]
N	3,591	3,591	3,591	460	460

5.11 Table 6: Competition and innovation

Read:

- In IFD industries, the interaction Competitive $\times$ Public is generally negative, indicating that product-market competition makes public firms innovate less than private firms.
- In EFD industries, competition does not generate the same negative innovation effect; if anything, R&D spending is higher for competitive public firms.
- This supports the idea that stock-market short-termism bites hardest when firms do not urgently need outside capital.

5.12 Table 7: Analyst coverage and innovation

Read:

- Analyst coverage is associated with lower future patents, citations, and generality overall.
- But the interaction with EFD is positive, meaning the negative analyst-coverage effect is much weaker for firms in EFD industries.
- Again, the paper's interpretation is not that analysts are harmless; it is that analyst-induced short-term pressure is especially damaging for firms that do not need public equity financing very much.

5.13 Table 8: Innovation efficiency

Read:

- Innovation efficiency is measured as patents per dollar of R&D.
- Public firms in EFD industries convert R&D into patents more efficiently than private firms; the public coefficient is insignificant in IFD industries.
- This matters because it shows the public-EFD advantage is not only higher spending. It is also more effective conversion of spending into innovation output.

Table 7

Analyst coverage and innovation.

This table reports the estimation results for the differential effects of analyst coverage on the innovation of public firms in external finance dependent (EFD) and internal finance dependent (IFD) industries. The following model is estimated: $Y_{i,k,t+3} = \alpha + \beta \ln(\text{Coverage})_{i,k,t} + \theta EFD_{i,k} \times \ln(\text{Coverage})_{i,k,t} + \gamma X_{i,k,t-1} + \lambda X_{i,k,t-1} \times EFD_{i,k} + \xi_i + \zeta_t + \varepsilon_{i,k,t}$, where the dependent variable is the innovation outcome measure including the three-year-ahead natural logarithm of one plus number of patents, the natural logarithm of one plus truncation bias-adjusted citations, originality, and generality. The independent variable $\ln(\text{Coverage})_{i,k,t}$ is the natural logarithm of one plus the average of the monthly numbers of earnings per share forecasts for firm i over fiscal year t from I/B/E/S. $EFD_{i,k}$ is external finance dependence dummy. $X_{i,k,t-1}$ is a set of firm characteristic variables that affect a firm's innovation outcome. ξ_i controls for firm fixed effects, and ζ_t captures year fixed effects. The coefficients on the control variables are not reported. Standard errors clustered at the firm level are reported in brackets. ***, **, and * indicate the 1%, 5%, and 10% significance level, respectively.

Variable	$\ln(\text{Patent})_{i,k,t+3}$	$\ln(\text{Citations})_{i,k,t+3}$	$\text{Originality}_{i,k,t+3}$	$\text{Generality}_{i,k,t+3}$
$\ln(\text{Coverage})$	-0.0648*** [0.0275]	-0.0906* [0.0501]	-0.0185 [0.0612]	-0.1570** [0.0787]
$EFD \times \ln(\text{Coverage})$	0.1029*** [0.0352]	0.1073* [0.0614]	0.0104 [0.0623]	0.1430* [0.0811]
N	14,272	14,272	5,284	5,284

Table 8

Innovation efficiency.

This table reports the estimation results for innovation efficiency of matched private and public firms in external finance dependent (EFD) and internal finance dependent industries (IFD). We estimate the treatment effect model to address the concern that a firm's decision to go public might not be random (selection bias). The treatment effect model is estimated with a two-step approach. In the first step, we estimate the probability of being public based on a firm's logarithm of total assets, capital expenditure, growth in sales, return on assets (ROA), and leverage from a probit model. The inverse Mills ratio (*Mills*) is included in the second-step to adjust for selection bias. The dependent variable is the innovation efficiency measured as natural logarithm of one plus patents per dollar research and development (R&D) investment [$\ln(1 + [\text{Patents}/\text{R\&D}])$]. The independent variables are the *Public* dummy; a set of characteristic variables that affect a firm's innovation activities, including $\ln(\text{Sales})$, *Tangible*, *Cash*, *Age*, *Capex*, growth in sales, and ROA; and year and industry fixed effects. In the last column, we estimate the treatment effect model with the second step model as $Y_{i,k,t} = \alpha + \beta \text{Public}_i + \delta EFD_{i,k} + \theta \text{Public}_i \times EFD_{i,k} + \gamma X_{i,k,t-1} + \lambda X_{i,k,t-1} \times EFD_{i,k} + \phi \text{Mills}_i + \varepsilon_{i,k,t}$, where $Y_{i,k,t}$ is innovation efficiency; and $EFD_{i,k}$ is an industry external finance index. $X_{i,k,t-1}$ is the set of control variables. Industry and time effects are included. The coefficients on the control variables are not reported. Two-step consistent standard errors are reported in the brackets. ***, **, and * indicate the 1%, 5%, and 10% significance level, respectively.

Variable	EFD industries	IFD industries	All
<i>Public</i>	0.0701*** [0.0129]	0.0066 [0.0164]	0.0176 [0.0142]
<i>EFD</i>			-0.0784 [0.0806]
<i>EFD</i> $\times$ <i>Public</i>			0.0669*** [0.0151]
<i>Mills</i>	-0.0265*** [0.0078]	-0.0007 [0.0101]	-0.0223*** [0.0068]
N	10,171	2,163	12,334

and *Public* dummy is positive and significant. The results indicate that public firms in EFD industries outperform private firms in innovation efficiency.

6.6. Acquisitions

Innovation can be achieved both internally and externally. [Seru \(2014\)](#) shows that innovation acquisition can be a more efficient way for mature firms with internal capital

6 Short summary

- **Conclusion.** Public listing helps innovation when firms need external capital, and does not help when firms can mostly fund investment internally.
- **Intuition.** For financially dependent firms, public markets relax financing constraints enough to dominate short-termism. For internally funded firms, the financing gain is smaller, so myopia can dominate.
- **Empirical lesson.** The strongest way to study public versus private firms is to combine a rich private-firm data set with multiple identification strategies: matching, treatment-effect correction, and a quasi-experimental delisting design.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The innovation effect of public listing is heterogeneous: it is positive for firms in EFD industries and close to zero for firms in IFD industries.
- **Magnitude.** In the treatment-effect estimates, public firms in EFD industries spend substantially more on R&D and produce more patents and citations; in the fuzzy-RD estimates, delisting sharply depresses future innovation only for EFD firms.
- **Credibility.** The main results survive matching, selection correction, placebo tests, density checks, pairwise EFD/IFD matching, and the exclusion of M&A acquirers.
- **Mechanism evidence.** Innovative EFD firms are more likely to go public, devote more IPO proceeds to R&D, are less prone to REM, and are less negatively affected by analyst coverage. Public firms in IFD industries appear more vulnerable to short-term pressure from competition and analysts.
- **Interpretation.** Public markets are not uniformly myopic or uniformly beneficial. Their net effect depends on whether access to equity financing relaxes a binding constraint.

7.2 Economic intuition

- Innovation is unusually hard to finance because it is intangible, risky, and informationally opaque. That makes public equity particularly valuable when internal funds are insufficient.
- Short-termism is real, but it is not the whole story. When the financing benefit is large enough, firms can still innovate more as public firms despite public-market pressure.
- The paper’s broad economic insight is therefore about a trade-off: *public markets provide both capital and pressure*. Which force dominates depends on the firm’s financial dependence.

7.3 Takeaway

This is a very useful paper for thinking about the real effects of stock markets. The right question is not whether public firms innovate more or less on average. The right question is which firms need the public market badly enough that financing benefits outweigh myopic pressure. Acharya and Xu

show that the answer is: firms in externally finance dependent industries. That makes the paper an important contribution to both the innovation literature and the broader debate over whether public equity markets encourage or discourage long-term investment.